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53. Sales Transaction Analysis

Files Required

- Sales.csv
- Stores.csv
- Sales.json
- Sales.xml

```
import pandas as pd

# Read CSV files
sales = pd.read_csv("sales.csv")
stores = pd.read_csv("stores.csv")

# Read JSON file
sales_json = pd.read_json("sales.json")

# Read XML file
sales_xml = pd.read_xml("sales.xml", parser="etree")

# Display input data
print("Sales Data:")
print(sales.head())

print("\nStores Data:")
print(stores.head())

print("\nSales Information:")
print(sales.info())

# Handle missing values
sales.fillna(0, inplace=True)

# Remove duplicate records
sales.drop_duplicates(inplace=True)

# Convert data type
sales["SalesAmount"] = sales["SalesAmount"].astype(float)
```

```
•
• # Remove extra spaces from column names
• sales.columns = sales.columns.str.strip()
• stores.columns = stores.columns.str.strip()
•
• # Merge Sales and Stores using StoreID
• merged = pd.merge(sales, stores, on="StoreID")
•
• # Display merged data
• print("\nMerged Data:")
• print(merged)
•
• # Summary statistics
• print("\nSummary Statistics:")
• print(merged.describe())
•
• # Sales by category
• summary = merged.groupby("Category")["SalesAmount"].sum()
•
• print("\nSales by Category:")
• print(summary)
•
• # Highest sales category
• print("\nHighest Sales Category:")
• print(summary.idxmax())
•
• # Export cleaned dataset
• merged.to_csv("Cleaned_Sales.csv", index=False)
•
• print("\nCleaned dataset exported successfully.")
```

Sample Data

Sales.csv

SaleID	StoreID	Category	SalesAmount
			t
1	101	Electronics	50000
2	102	Clothing	15000
3	101	Electronics	30000

4	103	Furniture	25000
---	-----	-----------	-------

Stores.csv

StoreID	StoreName
----------------	------------------

101	Store A
-----	---------

102	Store B
-----	---------

103	Store c
-----	---------

The screenshot shows a VS Code window with a Python script named 'query.py' being executed in a terminal. The script processes CSV files and outputs data for sales and stores. The output is as follows:

```
PS C:\Users\sandr\OneDrive\Desktop\Sales_Project> python query.py

Sales Data:
  SaleID  StoreID  Category  SalesAmount
0        1      101  Electronics      50000
1        2      102  Clothing       15000
2        3      101  Electronics     30000
3        4      103  Furniture      25000
4        5      104  Grocery       12000

Stores Data:
  StoreID  StoreName
0        101  Store A
1        102  Store B
2        103  Store C
3        104  Store D
4        105  Store E

Sales Information:
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   SaleID      10 non-null     int64
1   StoreID     10 non-null     int64
2   Category    10 non-null     str
3   SalesAmount 10 non-null     int64
dtypes: int64(3), str(1)
memory usage: 452.0 bytes
None

Merged Data:
  SaleID  StoreID  Category  SalesAmount  StoreName
0        1      101  Electronics      50000.0  Store A
1        2      102  Clothing       15000.0  Store B
2        3      101  Electronics     30000.0  Store A
3        4      103  Furniture      25000.0  Store C
4        5      104  Grocery       12000.0  Store D
5        6      105  Electronics     45000.0  Store E
6        7      102  Clothing       18000.0  Store B
7        8      103  Furniture      32000.0  Store C
8        9      104  Grocery       14000.0  Store D
9       10      105  Electronics     60000.0  Store E

Summary Statistics:
  SaleID  StoreID  SalesAmount
count    10.000000    10.000000    10.000000
mean      5.500000    103.000000   30100.000000
std        3.027655     1.490712   16676.330532
min        1.000000    101.000000   12000.000000
25%        3.250000    102.000000   15750.000000
50%        5.500000    103.000000   27500.000000
75%        7.750000    104.000000   41750.000000
max       10.000000    105.000000   60000.000000

Sales by Category:
Category
Clothing      33000.0
Electronics  185000.0
Furniture     57000.0
Grocery       26000.0
Name: SalesAmount, dtype: float64

Highest Sales Category:
Electronics

Cleaned dataset exported successfully.
PS C:\Users\sandr\OneDrive\Desktop\Sales_Project>
```

54. Hospital Patient Records

Files Required

- Patients.csv
- Doctors.csv

- Patients.json
- Patients.xml

```
import pandas as pd

# Read CSV files
patients = pd.read_csv("patients.csv")
doctors = pd.read_csv("doctors.csv")

# Read JSON file
patients_json = pd.read_json("patients.json")

# Read XML file
patients_xml = pd.read_xml("patients.xml", parser="etree")

# Inspect dataset
print("Patients Data:")
print(patients.head())

print("\nDataset Information:")
print(patients.info())

# Handle missing values
patients.fillna("Unknown", inplace=True)

# Remove duplicate records
patients.drop_duplicates(inplace=True)

# Validate data types
patients["Age"] = patients["Age"].astype(int)

# Merge Patients and Doctors using DoctorID
merged = pd.merge(patients, doctors, on="DoctorID")

print("\nMerged Data:")
print(merged)

# Summary statistics
print("\nSummary Statistics:")
print(merged.describe())

# Aggregation
summary = merged.groupby("Department")["Age"].mean()

print("\nAverage Age by Department:")
```

```

• print(summary)
•
• # Department with highest average age
• print("\nDepartment with Highest Average Age:")
• print(summary.idxmax())
•
• # Export cleaned dataset
• merged.to_csv("Cleaned_Patients.csv", index=False)
•
• print("\nCleaned dataset exported successfully.")

```

Sample Data

Patients.csv

PatientID	Name	Age	DoctorID
1	Rahul	25	101
2	Anu	35	102
3	Asha	40	101
4	John	50	103

Doctors.csv

DoctorID	DoctorName	Department
101	Dr. Ravi	Cardiology
102	Dr. Meena	Neurology
103	Dr. Arun	Orthopedics

File Edit Selection View Go Run Terminal Help

← → Hospital_Project

Update

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SESSIONS

EXPLORER

HOSPITAL_PROJECT

Cleaned_Patients.csv

doctors.csv

patients.csv

patients.json

patients.xml

query.py

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\sandr\OneDrive\Desktop\Hospital_Project> python query.py

Patients Data:

PatientID	Name	Age	DoctorID	
0	1	Rahul	25	101
1	2	Anu	35	102
2	3	Asha	40	101
3	4	John	50	103
4	5	Priya	28	102

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 4 columns):

#	Column	Non-Null Count	Dtype
0	PatientID	10 non-null	int64
1	Name	10 non-null	str
2	Age	10 non-null	int64
3	DoctorID	10 non-null	int64

dtypes: int64(3), str(1)

memory usage: 452.0 bytes

None

Merged Data:

PatientID	Name	Age	DoctorID	DoctorName	Department	
0	1	Rahul	25	101	Dr. Ravi	Cardiology
1	2	Anu	35	102	Dr. Meena	Neurology
2	3	Asha	40	101	Dr. Ravi	Cardiology
3	4	John	50	103	Dr. Arun	Orthopedics
4	5	Priya	28	102	Dr. Meena	Neurology
5	6	Arun	45	103	Dr. Arun	Orthopedics
6	7	Meera	32	101	Dr. Ravi	Cardiology
7	8	Ravi	55	103	Dr. Arun	Orthopedics
8	9	Sneha	29	102	Dr. Meena	Neurology
9	10	Kiran	38	101	Dr. Ravi	Cardiology

Summary Statistics:

	PatientID	Age	DoctorID
count	10.00000	10.000000	10.000000
mean	5.50000	37.700000	101.900000
std	3.02765	9.888826	0.875595

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Update

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SESSIONS

EXPLORER

HOSPITAL_PROJECT

Cleaned_Patients.csv

doctors.csv

patients.csv

patients.json

patients.xml

query.py

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Summary Statistics:

	PatientID	Age	DoctorID
count	10.00000	10.000000	10.000000
mean	5.50000	37.700000	101.900000
std	3.02765	9.888826	0.875595
min	1.00000	25.000000	101.000000
25%	3.25000	29.750000	101.000000
50%	5.50000	36.500000	102.000000
75%	7.75000	43.750000	102.750000
max	10.00000	55.000000	103.000000

Average Age by Department:

Department

Cardiology	33.750000
Neurology	30.666667
Orthopedics	50.000000

Name: Age, dtype: float64

Department with Highest Average Age:

Orthopedics

Cleaned dataset exported successfully.

PS C:\Users\sandr\OneDrive\Desktop\Hospital_Project> |

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Q Search

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